Getting into the technical aspects of the Cell Processor would entail a lesson in microprocessor technology, with phrases like 'Power Processor Element' and 'Synergistic Processing Elements' being thrown about every couple of sentences.

In layman's terms, the Cell Processor is an eight-core processor, which means that it can compute eight different processes at the same time – a fantastic achievement to improve the speed of computing. Each of these eight 'cores' is also fitted with 256 KB of memory for local storage when processing an instruction.

The biggest problem for IBM with building such a chip was in figuring out how to use each core to the fullest. The introduction of a 'Resource Allocation' algorithm solved this problem by ensuring that the PS3 is smart enough to determine which cores are being used at a time and allot new operations accordingly.

In the end, the 64-bit architecture, coupled with the 3.2 GHz core clock speed and the 256x8KB of storage, translates to some awe-inspiring computational speeds.

The Cell Processor, in theory, can be over-clocked to 4 GHz quite easily, but since the PS3 runs at 3.2 GHz, that is the most commonly cited clock speed for it.

A worrying factor for such a high-powered processor is over-heating. In the PS3, Sony managed to make room for a high-quality heat-sink that enables the Cell Processor to run smoothly at temperatures as high as 85 degrees Celsius.

Now, with these kinds of speeds, the amount of reliance on other factors of computing, such as the RAM or the graphics card, is substantially reduced. Not that it stopped Sony from coming out with a jaw-dropping graphics chipset…

RSX Reality Synthesizer

Just as Sony went to the leaders of the chip-making world for its Cell Microprocessor, it went to the leaders of the graphics processing unit (GPU) manufacturers for the PS3's graphics card. Nvidia was glad to help out in building what Sony eventually called the 'Reality Synthesizer RSX' graphics unit.

More powerful than two GeForce 6800 Ultra cards put together (and that's together, not SLI)

At that time, Nvidia was busy preparing for its own 7000-series of graphics card and had already come out with the G70 base architecture for its premium model, the GeForce 7800GT. Sony liked what it saw and went with that – not a future-proof graphics card, for sure, but the company didn't feel the